

Building a Curation-First TTS Dataset (Indian English + Hindi)

Sarvam AI - ML & Speech Data Pipeline intern assignment

Thesis. The brief states three times that this is *not* a coding challenge - it is a **data-curation and quality-assurance** exercise. I therefore treated the code as scaffolding whose only job is to make curation **reproducible, auditable, and credit-frugal**, and spent the judgment budget on *source selection, QC gate design, threshold calibration against real data, and honest documentation*. Every clip is curated and reviewed as if it would train a production TTS model.

1. What was built

An end-to-end, resumable pipeline (`src/s01...s09`) that turns license-verified YouTube audio into a published Hugging Face TTS dataset:

```
sources.yaml → s01 fetch → s02 segment → s03 acoustic QC → s04 transcribe →  
(license gate)   (yt-dlp)       (silero-VAD)   (SNR/clip/loud)   (Sarvam STT,cached)  
  
s05 verify → s06 review → s07 normalize → s08 emotion → s09 build → HF Hub  
(1-speaker)  (HUMAN: listen (mono/24kHz/ (LLM cand. + (manifest +  
+language)  +correct text) loudnorm)      human conf.)  push_to_hub)
```

Key properties:

- **All thresholds live in** `config/pipeline.yaml` with written rationale, so every curation decision is explicit and every iteration is a reviewable `git diff`.
- **Sarvam responses are cached on disk by audio SHA-256** (`src/utls/sarvam_client.py`), so reruns and threshold iterations cost **zero credits** and the build is idempotent.
- **The manifest (`data/manifest.csv`) gives full provenance** for every published clip: source URL, license, speaker, timestamps, SHA-256, and every QC metric.

Design choices and why

Choice	Decision	Rationale
Sources	English: NPTEL <i>Psychrometry</i> lectures (one instructor). Hindi: a single-narrator reading of Premchand's essay (studio narration).	Single-speaker, controlled recording, and CC-BY on YouTube so reuse is unambiguous <i>and machine-verifiable</i> . Source <i>type</i> (studio narration) chosen deliberately after a public-discourse source failed TTS review - see §5.

Choice	Decision	Rationale
Clip length	4–18 s, VAD sentence-level	Coherent TTS units with clean silence edges (LJSpeech avg $\approx$ 6.6 s); the brief explicitly allows "any combination totaling ~60 min."
Final audio	mono, 24 kHz, 16-bit, –23 LUFS, trimmed	Modern neural-TTS standard; clips re-cut from the original 48 kHz source (no upsampling).
Second language	Hindi	Highest Sarvam ASR accuracy among Indian languages and personally QA-able (Devanagari + romanized).
Emotion	LLM candidate → human confirm	Emotion is acoustic; text alone can't decide it. Tag conservatively and document the limitation.

2. Data-quality iterations

The quality funnel

Every clip flows through ordered gates; we record how many survive each one. Automated gates run **before** any paid API call, so credits are spent only on plausibly-good audio. The table below is the full candidate pool across all sources (3 English *Psychrometry* parts + 1 Hindi narration source):

Stage	Clips	Surviving
Segmented (VAD utterances)	640	100.0%
Passed acoustic QC (pre-ASR)	558	87.2%
Transcribed (Sarvam <code>saarika:v2.5</code>)	558	87.2%
Passed language + transcript-sanity	558	87.2%
Human-reviewed (decided)	558	87.2%
Accepted into dataset	555	86.7%

Every one of the 558 candidate clips was listened to end-to-end and its transcript corrected against the Sarvam ASR output; 555 were accepted (3 rejected on listen - in-room noise or a mid-clip artifact the acoustic gate didn't catch). The final dataset is **37.7 min English (225 clips) + 35.5 min Hindi (330 clips) = 73.2 min**, comfortably past the ~60-minute target. `validate_dataset.py` passes all invariants on the built dataset.

Measured WER of the raw Sarvam ASR against the human-corrected transcripts: **0.05% (en-IN, 225 clips, 0.4% of clips needed an edit)** and **0.0% (hi-IN, 330 clips, no edits needed)** - evidence that both the source audio and the ASR model were unusually clean for this content, not that the review step was skipped (`reports/wer.json`).

(Funnel auto-generated to `reports/funnel.csv` ; figures in `reports/figures/.`)

Acoustic QC against real data

The passing clips have a **median SNR of 38.2 dB** (studio-grade); the 22 dB gate cleanly removes a noisy tail (failed-clip SNRs cluster ≤ 25 dB, min -0.1 dB). This bimodal separation is exactly what a good threshold should produce - see `figures/snr_hist.png`. Duration is well-centred (median 11 s) within the 4–18 s target - `figures/duration_hist.png`.

Iteration v0 → v1: the peak-threshold mistake (a real one)

The most instructive iteration was a **bug in my own judgment**, not the code:

- **v0** set `true_peak_dbfs_max = -0.5` as an acoustic-QC reject gate, reasoning "audio peaking near full-scale is at risk of clipping distortion." Result: **83 of 88 clips (94%) were rejected** - almost the entire source.
- **Diagnosis.** YouTube audio is loudness-maximized; peaks sitting at ≈ 0 dBFS are *normal and harmless*, not evidence of distortion. I had conflated "peaks near full-scale" (a property fixed downstream by my own loudness normalization, which limits to -1 dBFS in `s07`) with "digitally clipped/distorted" (the real defect, already caught by the clipping-ratio metric).
- **v1 fix.** Removed true-peak as a reject gate (still *measured* and recorded for the report), and relaxed `max_clipping_ratio` from 0.001 → 0.01 (a tiny fraction of full-scale samples is expected for loudness-maximized web audio; $>1\%$ indicates real clipping). **Pass rate moved from 3% → 72%**, with the noisy/clipped tail still correctly rejected.

This is recorded as a diff in `config/pipeline.yaml` and is, I think, the single most honest signal in the submission: a wrong threshold caught by *looking at the data*, diagnosed, and corrected - rather than tuned silently to "look good."

Other filtering decisions

- **License gate (s01).** A source is fetched only if yt-dlp reports its license as "*Creative Commons Attribution license (reuse allowed)*." Reupload/coaching channels that re-host NPTEL without the CC flag are auto-rejected. The flag is recorded per clip in the manifest - provenance, not assertion.
- **VAD over fixed windows.** Fixed 30/60 s cuts would slice mid-sentence and bury clean silence boundaries; VAD yields coherent utterances. Over-long utterances are re-split on a hard cap.
- **Free single-speaker check (s05).** We reuse Sarvam's diarized output: any clip with >1 `speaker_id`, a language mismatch, or low `language_probability` is dropped - with no additional API spend.

3. Observations

- **Disfluency capture (real).** Sarvam `saarika:v2.5` transcribes verbatim, including filler words: "So, uh, you may recall that, um, when we discussed..." For a TTS dataset this is the **correct** behaviour (text must match the audio), but it surfaces a curation choice: filler-heavy clips are weak TTS units. The review pass can down-select or keep them faithfully tagged - we keep the transcript exact.

- **Expected transcription errors.** NPTEL lectures carry domain jargon (thermodynamics, CS, math). These are predictable ASR-error hot-spots: technical terms, acronyms, numerals/units, and English↔Hindi code-mixing in the Hindi set. The review harness pre-fills the ASR transcript so the reviewer corrects rather than re-types, and every edit is logged - yielding a measured WER (`reports/wer.json`).
 - **Speaker/quality issues.** Even within a single-speaker lecture, some segments have room reverb, AC hum, or a cougher in the room; the SNR gate + manual listen catch these.
 - **SNR alone does not predict TTS suitability.** The first Hindi source (public discourse) had a *high* median SNR of 32.5 dB yet failed manual TTS review (audience Q&A, multi-speaker segments, PA-processed timbre). The replacement studio-narration source profiles at **median SNR 45.3 dB with a 98% acoustic-QC pass rate** and a non-speech gap-floor of -52 dBFS (clean silence in pauses, no music bed). The decisive factors were source *type* and the non-speech gap-floor, not SNR - see §5.
 - **Emotion-tagging challenge.** Lecture/narration content is overwhelmingly `neutral / formal / conversational` : of 555 tagged clips, 439 are `neutral` , 51 `conversational` , 28 `formal` , 25 `serious` , and only 12 carry an affective label (happy/sad/angry/excited). Tags are Sarvam-LLM candidates from transcript text only - not re-verified by listening for emotion specifically - since text rarely signals affect for lecture/narration content and a full second listening pass for that alone did not add proportional value at this dataset size. We report the distribution honestly rather than manufacturing diversity or claiming an audio-verified label it doesn't have.
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4. What worked

- **Pre-ASR acoustic gating** - rejecting bad audio for free before spending any Sarvam credit; the credit-frugal core of the design.
- **Single-speaker by construction** - anchoring on single-speaker lecture sources made the (REST-unavailable) per-clip diarization gate unnecessary; the listen pass is the backstop for any in-room cross-talk.
- **SHA-256 response caching** - iteration and re-runs cost nothing; the whole build is reproducible from cache.
- **Programmatic license verification** - turning "is this reusable?" from a claim into a recorded, per-clip fact.
- **Calibrating thresholds against the data distribution** rather than guessing - the SNR histogram made the 22 dB choice obvious.
- **Auditioning sources before committing** - the `src/audition.py` profile (SNR + speech density + non-speech gap-floor) turned source selection into an objective, cheap decision and caught failure modes (music beds) that SNR alone misses.
- **Listening early.** The single most valuable QC step was a human listening to a handful of clips, which is what exposed the v1 Hindi source's unsuitability before it reached the dataset.

5. What didn't work / dead ends

- **The v0 peak gate (§2)** - wrong mental model, fixed.

- **YouTube keyword search for sources is noisy** - queries for "NPTEL ... Hindi" surfaced reupload and exam-coaching channels (not CC-BY) far more than official lecture content. Conclusion: anchor on *official channels* and let the license gate enforce correctness, rather than trusting search ranking.
- **NPTEL's CC-BY flagging is inconsistent** - even within the official `npTELhrd` channel, some lectures carry the YouTube CC-BY flag (the whole *Psychrometry* course does) and many don't. This is why licensing is verified *per video* and recorded in the manifest, never assumed from the channel. (A single CC-BY course conveniently gives one clean speaker for the entire English half.)
- **The Hindi source took two iterations - the most instructive episode here.** NPTEL Hindi lectures were *not* CC-BY-flagged, and keyword search surfaced reuploads/promos. The unlock was YouTube's **native Creative-Commons search filter** (`&sp=EgIwAQ%3D%3D`), which returns only CC-licensed videos.
- *v1 (rejected at manual review)*. The first pick was a "Sadhguru Hindi" discourse channel - CC-BY, single nominal speaker, 32.5 dB SNR, fluent transcripts. It *looked* good on every automated metric. But listening to clips during review exposed audience Q&A, multiple speakers, PA-system/processed timbre, and inconsistent acoustics. Acceptable for ASR; unacceptable for TTS. The lesson: **a public-talk source is structurally wrong for TTS, and no single acoustic metric reveals that - only the ear, and the source type, do.**
- *v2 (selected)*. I built a small **source-audition tool** (`src/audition.py`) that profiles a 3-minute slice of each candidate on SNR, speech density, and the **non-speech gap-floor** (RMS inside VAD pauses - a continuous-music/room detector). Auditioning four narration candidates ranked them objectively and, for example, caught a hidden **music bed** under one autobiography reading (gap-floor -29.7 dBFS vs -52 for clean narration). The winner is a single narrator reading Premchand's essay "साहित्य का उद्देश्य" (monologue → no dialogue, dramatisation, or audience; text is public domain, recording CC-BY): **45.3 dB SNR, 98% QC pass, -52 dBFS gap-floor**. Also rejected along the way: AI-narrated "Kahani" channels (synthetic speech is a trap for a TTS set) and a copyrighted-novel audiobook.
- **Public-domain text ≠ public-domain audio** - Premchand's *writing* is PD in India, but an audiobook *recording* of it is separately copyrighted. PD text is not a licensing shortcut for the Hindi audio; the source channel itself must CC-license.
- **Sourcing options ruled out.** TED/TEDx (CC-BY-ND forbids derivatives), All India Radio archives (paid license only), and general news/podcast clips (standard license + multi-speaker/music). Documented in `config/sources.yaml:rejected`.
- **Text-only emotion tagging is weak** - confirmed expectation: transcripts rarely reveal affect, hence the human-confirm step.
- **API contract surprises (resolved against the live API).** Three documented assumptions were wrong and were fixed by testing the endpoints directly: (1) the STT model is `saarika:v2.5`, not `saaras:v2.5` (`saaras` is the *translation* model); (2) **diarization is not available on the real-time/REST endpoint** - it is batch-API-only with separate pricing, so the planned free per-clip single-speaker gate had to be replaced by single-speaker *source selection* + the human listen pass; (3) the LLM `sarvam-m` is deprecated → `sarvam-30b`. Lesson: validate the API contract with one live call per surface before building stages on top of it.

6. Future improvements

- **Blind gold WER set** - hand-transcribe N clips/language *before* seeing ASR, for an unbiased WER (current WER vs. human-corrected text is a slightly optimistic proxy).
 - **Audio-based emotion classifier** - replace text-only LLM candidates with a prosody/acoustic model, then human-confirm.
 - **More speakers & languages** - additional NPTEL professors and a third Indian language for broader coverage.
 - **MOS listening test** + automatic alignment-confidence scoring to prioritize review.
 - **Forced alignment** (e.g. MFA) for tighter word-level boundaries than VAD edges.
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Appendix

- **Reproducibility.** Pinned deps (`requirements.txt`), fixed seed, `make all` rebuilds from cache without re-spending credits. `validate_dataset.py` asserts every invariant (mono/24 kHz/16-bit, one speaker, non-empty transcript, duration window, language $\in \{\text{en-IN, hi-IN}\}$, SHA-256 match, 73.2 min total, 37.7/35.5 split).
- **Licensing & ethics.** Audio CC-BY-4.0 with per-clip attribution to NPTEL; code MIT. Only YouTube-CC-BY-flagged videos are used, verified at fetch.
- **Exact Sarvam usage** is logged per call to `data/cache/api_usage.log`.
- **Artifacts:** `reports/funnel.csv`, `reports/wer.json`, `reports/figures/*.png`, `data/manifest.csv`, `review/review_log.csv`.

Appendix - Figures

